

BINLEVEL · REQUIREMENTS

BinLevel requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-30	Amish Chadha	Design for construction (BNL-DDR-003); R1, R14 and R16 figures updated from BNL-CAL-001 v0.3; no status changed
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Status checked by calculation (BNL-CAL-001); adopted defaults from BNL-DDR-001 written into R3 and R9; 11-byte payload
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

BinLevel requirements

These requirements are the TRL 3 design basis. The design choices behind them were decided by Amish on 2026-09-25 (go with recommendation; BNL-DDR-001 and BNL-DDR-002). Under DDR-002, R6's upper limit is raised from 60 to 70 °C, R9's rate-of-rise rule counts only above 50 °C, and R3's 2 h interval applies at SF12 only; no other target changed. Under BNL-DDR-003 (design for construction, 2026-09-30, open for Amish's review) the box grows to 55 mm and gains studs, standoffs and printed sensor mounts; the mass, envelope, geometry and cost figures below are updated and no requirement changed status. Status is from the sizing note BNL-CAL-001 v0.3, which checks each requirement against the parametric model: two are not met (R5, R10 in steel containers), two are at risk (R2, R6), two can only be settled by test (R7, R12) and ten are met (six by calculation, four by design). See the notes under Table 1.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (BNL-CAL-001)
R1	Measure distance from lid to waste surface over the full depth of the target container	0.03 to 1.5 m below the sensor face	Calculation and datasheet review; later bench test in a container	Met: 0 to 1.00 m needed; ultrasonic from 0.25 m, ToF to 0.4 m, overlapping between 60 % and 75 % fill (datasheet ranges to confirm)
R2	Report fill level accurately on real mixed waste	Within ± 10 % of container depth for 90 % of readings	Error budget; later field comparison against manual dip readings	At risk: instrument error about 10 mm (ultrasonic) and 20 mm (ToF) of the ± 101 mm allowance; the waste surface is unknown
R3	Report often enough to plan routes, and quickly when full	Routine uplink at least every 2 h (default 1 h, 2 h at SF12 only); threshold alert within 20 min of crossing the set fill level (default 80 %)	Timing calculation	Met: worst-case alert 17.5 min
R4	Run for years on one battery	5 years or more in the worst radio case (SF12); 10 years design target	Power budget calculation	Met: 16.3 years on a C cell at SF12; enclosure and seals, not the cell, limit life
R5	Survive rain, condensation and washing	IP67 minimum; IP69K target where containers are pressure washed	Enclosure datasheet; later spray test	Not met for IP69K: stock IP67 box (DDR-001, D9)
R6	Operate across the climate range	-20 to +70 °C inside the container (raised from +60 °C, DDR-002); RAK3172-T for sites colder than -20 °C	Heat balance and datasheet review	At risk: a dark lid reaches about 67 °C on a 40 °C day, 3 K inside the limit; the standard module is rated only down to -20 °C; transducer rating and condensation unverified
R7	Survive lifting, tipping and lid slams	Repeated shocks of the emptying cycle without loosening; target IK08 impact on the enclosure	Load calculation; later drop and shake test	Not verifiable at TRL 3: 85 N at 20 g against about 4,200 N per bolt; loosening and impact need a test
R8	Detect and report emptying and lid events	Tip event from the accelerometer reported with the next uplink	Firmware logic review	Met by design
R9	Warn of abnormal heat in the bin	Alert uplink within 15 min when temperature exceeds 70 °C, or rises more than 15 K in 15 min while above 50 °C (gate added, DDR-002)	Timing calculation and heat balance	Met on paper: alert within 7.5 min with temperature read every 5 min; sun through cloud starts the unit at about 43 °C, below the gate, and above it the worst sun rise is 10.9 K; maintenance aid only, not a fire detector
R10	Connect over open LoRaWAN within radio rules	LoRaWAN 1.0.x Class A; EU868, US915 and IN865 plans; within duty cycle and TTN's 30 s per day fair use	Airtime and link calculation; later range test	Not met in steel containers with the internal antenna (about 0.35 km at SF12); met in HDPE containers: 20.8 s per day at SF12, 11-byte payload fits US915 DR0, about 0.8 to 1.3 km range

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (BNL-CAL-001)
R11	Protect privacy	No camera or microphone; payload limited to fill, distance, temperature, tilt, events and battery	Design review of BOM and payload	Met by design
R12	Install without wiring or tools beyond a drill	Fit to a lid in 10 min or less with four bolts or rivets; no cables	Timed trial at TRL 4	Not verifiable at TRL 3: four through-bolts on a 130 x 60 mm pattern, no cables
R13	Resist casual tampering and theft	Mounted inside the lid; tamper-resistant fasteners	Design review	Met by design
R14	Low cost and buildable	Parts cost \$60 or less per sensor at prototype quantities; no custom tooling	Priced BOM	Met: \$59.00 (indicative, \$1.00 margin, BNL-DDR-003); \$67.00 with the external antenna option
R15	Open and interoperable	Documented payload format and open decoder; works with any LoRaWAN network server	Documentation review	Met by design: candidate 11-byte layout in BNL-CAL-001
R16	Light and compact	400 g or less; no larger than 160 x 90 x 100 mm below the lid, including bracket	Mass calculation from the model, then weighing	Met: about 350 g with a 2 mm aluminium plate and the 55 mm box (DDR-003; 334 g before, 435 g with 1.5 mm stainless); 150 x 80 x 71.5 mm

Notes on requirements not met or at risk:

- **R2 at risk.** Ultrasonic echoes from bags, boxes and bulky items are uneven, and the waste surface is rarely flat. The mitigation is to take the median of several pings and to combine the two sensors, but accuracy on real waste is unknown until measured.
- **R5 not met for IP69K.** A low-cost stock enclosure is typically rated IP67. Where containers are washed with hot pressure water, a sealed potted design or an IP69K enclosure is needed, which raises cost.
- **R6 at risk.** The upper limit is now 70 °C (DDR-002), above the about 67 °C a dark lid reaches in hot sun, but the margin is only 3 K and the transducer's rating is unconfirmed. The RAK3172-T (-40 to 85 °C) is named for cold sites.
- **R10 not met in steel containers.** An antenna inside a steel container loses about 30 dB (assumed). HDPE containers are fine within about 1 km of a gateway. The external antenna or the exclusion of steel containers is still open, awaiting Amish (DDR-001, O1).

■ Assumptions

- Target container: 1,100 L four-wheel communal container (DDR-001, D1), lid underside 1.07 m above the inner floor and transducer face 1.00 m above it (from the parametric model).
- Fill level is reported as a percentage of the distance from the sensor face to the container floor, measured at installation.

- Worst radio case is spreading factor 12 at 125 kHz with an 11-byte payload (1.48 s airtime per uplink); typical urban case is SF9 (0.21 s).
- A dark lid may reach about 67 °C on a 40 °C day in full sun (BNL-CAL-001, section H; to be measured).